

Statistical and Machine Learning

Exam (Mock version)

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Name:

Student ID:

Instructions. This is a partially open-book exam. The following auxiliary materials are allowed:

- a non-programmable calculator;
- one A4 page of handwritten notes, written on both sides.

Notes written on tablets are not allowed. Bringing or using tablet-written or printed notes will be considered cheating and will be penalised with a grade of 5.

Exercise 1 — True / False (20 marks)

For each of the ten statements below, (i) decide whether it is **True** or **False** (1 mark), and (ii) give a brief justification (1 mark).

(a). (Linear models) Consider the ordinary least square regression with parameter $\beta \in \mathbb{R}^p$ estimated by $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ such that $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \mathbf{H}\mathbf{Y}$.

Statement: $\mathbf{H}$ is idempotent, i.e., $\mathbf{H}\mathbf{H} = \mathbf{H}$.

Circle one: **True** / **False**

Justification:

(b). (Evaluation) Consider the classification problem. The ROC curve plots the False Positive Rate on the x -axis against the True Positive Rate on the y -axis.

Statement: If the AUC (area under the curve) is 0.75, the classifier is worse than the random guessing.

Circle one: **True** / **False**

Justification:

(c). **(Regularisation & Bayesian)** For $\lambda > 0$, the ridge estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$ exists even when $\mathbf{X}^\top \mathbf{X}$ is singular.

Circle one: **True** / **False**

Justification:

(d). **(Exemplar & kernel methods)** k -nearest-neighbour classification is a parametric method that assumes a specific functional form for the decision boundary.

Circle one: **True** / **False**

Justification:

(e). **(Trees & boosting)** In a regression tree, the prediction assigned to a terminal node (leaf) is the mean of the response values of the training observations falling in that node.

Circle one: **True** / **False**

Justification:

(f). **(Clustering)** Purity is a reliable measure of clustering quality because it penalises solutions that use a very large number of clusters.

Circle one: **True** / **False**

Justification:

(g). **(Dimension reduction)** In PCA, the first principal component is the direction that maximises the variance of the projected data.

Circle one: **True** / **False**

Justification:

(h). **(Neural networks — tabular)** A multilayer perceptron using only linear activation functions can represent a strictly larger class of functions than a single linear model.

Circle one: **True** / **False**

Justification:

(i). (**Neural networks — images**) A convolutional layer typically has far fewer parameters than a fully connected layer applied to the same image, because its filter weights are small and shared across spatial locations.

Circle one: **True** / **False**

Justification:

(j). (*Neural networks — sequences*) In a standard RNN, the hidden state $\mathbf{h}_t$ depends only on the current input $\mathbf{x}_t$ and not on the previous hidden state $\mathbf{h}_{t-1}$.

Circle one: **True** / **False**

Justification:

Exercise 2. Linear models (5 marks)

(a) (3 marks) Consider the ordinary least square regression with parameter $\beta \in \mathbb{R}^p$ estimated by $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. Under the Gauss–Markov assumptions $\mathbb{E}(\varepsilon) = \mathbf{0}$ and $\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}_n$, derive $\mathbb{E}(\hat{\beta})$ and $\text{Cov}(\hat{\beta})$.

(b) (2 marks) Give two reasons why a standard linear regression model is inappropriate for modelling if the outcome variable is binary, i.e. $Y \in \{0, 1\}$.

Question 3. Evaluation (5 marks)

(a) (3 marks) A binary classifier gives $TP = 40$, $FP = 10$, $FN = 20$, $TN = 30$. Compute the accuracy and the precision and give one example of situation where precision would be an important metric to report.

(b) (2 marks) Consider a K-fold cross-validation. Give one method to select the number of folds and explain what is the purpose of holding one fold at a time?

Question 4. Regularisation & Bayesian (5 marks)

(a) (2 marks) Write the ridge-regression penalised least-squares objective $PLS(\beta)$ and give the closed-form ridge estimator.

(b) (3 marks) For orthonormal covariates ($\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$), show that ridge scales each OLS coefficient by $\frac{1}{1+\lambda}$, and describe the limits $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.

Question 5. Exemplar & kernel methods (5 marks)

(a) (2 marks) Write the k -NN estimate of the probability that a point $\mathbf{x}$ belongs to class l , and define every symbol.

(b) (3 marks) For $k = 5$, a query point has nearest neighbours with labels $\{A, A, B, A, C\}$. Give the predicted class and $\hat{P}(Y = A \mid \mathbf{x})$, then state one advantage and one disadvantage of k -NN relative to logistic regression.

Question 6. Trees & boosting (5 marks)

(a) (3 marks) A two-class node contains 10 observations: 6 of class A and 4 of class B ($\hat{\pi}_A = 0.6$, $\hat{\pi}_B = 0.4$). Compute:

- b) the misclassification error $i_1(R_m) = 1 - \max_k \hat{\pi}_{mk}$,
- c) the Gini index $i_2(R_m) = \sum_{k=1}^K \hat{\pi}_{mk}(1 - \hat{\pi}_{mk}) = 1 - \sum_{k=1}^K \hat{\pi}_{mk}^2$ where $k = 1, \dots, K$ stands for the class index, and R_m stands for terminal region.

(b) (1 mark) What do the above metrics measure?

Question 7. Clustering (5 marks)

(a) (3 marks) Describe the two steps of the K -means algorithm (assignment and update) with formulas, and state the objective (distortion) it minimises.

(b) (2 marks) Explain why the number of clusters K cannot be chosen by minimising the distortion J on the data, and name one principled alternative.

Question 8. Dimension reduction (5 marks)

(a) (3 marks) Explain the encode/decode view of PCA and write the reconstruction error (distortion) being minimised; state what the optimal $\mathbf{W}$ is in terms of the covariance matrix.

(b) (2 marks) State the equivalence between minimising reconstruction error and maximising variance, and hence what the first principal component maximises.

Question 9. Neural networks — tabular (5 marks)

(a) (2 marks) Explain why an MLP needs nonlinear activation functions: what happens if every activation is linear?

(b) (3 marks) Give the sigmoid $\sigma(a)$ and its derivative, explain the vanishing-gradient problem it causes, and name one activation that mitigates it and why.

Question 10. Neural networks — images (5 marks)

(a) (3 marks) Give the output-size formula (one spatial dimension) for a convolution with input size x , filter size f , zero-padding p per side and stride s ; then compute the output size for $x = 7$, $f = 3$, $p = 1$, $s = 2$.

(b) (2 marks) In what sense is a convolutional layer a special case of a fully connected (MLP) layer? Give the two structural properties of its weight matrix and what they buy you.

Question 11. Neural networks — sequences (5 marks)

(a) (3 marks) Write the RNN hidden-state update $\mathbf{h}_t$ (in terms of the input, previous output, and previous hidden state) and explain why $\mathbf{y}_t$ implicitly depends on all past observations.

(b) (2 marks) What is backpropagation through time (BPTT), and why is truncated BPTT commonly used?

Question 12. Linear models, Regularisation & Bayesian, Evaluation (10 marks)

(a) (6 marks) Consider ridge regression with tuning parameter λ . (i) Using the bias–variance decomposition, explain how increasing λ affects the bias and the variance of $\hat{\boldsymbol{\beta}}_{\text{ridge}}$, and describe the two limiting cases $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$. (3 marks) (ii) Describe how K -fold cross-validation is used to select λ in practice, and explain why choosing λ by minimising the *training* error alone would always lead to selecting $\lambda = 0$ (3 marks).

(b) (4 marks) Give a real-world regression scenario with many correlated predictors (e.g. gene-expression or economic data) and explain, using the bias–variance trade-off, why you would prefer a regularised approach over plain OLS. If your goal is also automatic variable selection, which regulariser would you choose, and why?

Question 13. Trees & boosting, Neural networks (10 marks)

(a) (6 marks) (i) Random forests reduce variance by averaging many de-correlated trees; deep neural networks are often regularised using dropout. Explain the conceptual similarity between these two techniques as implicit forms of model averaging. (ii) Give one structural reason a fully connected deep neural network may overfit more easily than a tree ensemble on a tabular dataset with only a few thousand observations.

(b) (4 marks) You are asked to build a fraud-detection model on a tabular dataset with mixed numeric and categorical features and only around 5,000 labelled examples. Would you recommend a gradient-boosted tree ensemble or a deep MLP as a first approach? Justify your choice with reference to sample size, interpretability, and regularisation.

Question 14. Dimension reduction, Clustering (10 marks)

(a) (6 marks) (i) Explain why applying PCA before running K -means on high-dimensional data is often beneficial, referencing both the reconstruction-error view of PCA and the distortion objective of K -means. (ii) The Gaussian mixture model (GMM) fitted by EM can be viewed as a probabilistic generalisation of K -means. State one assumption on the mixture components under which the GMM's E-step responsibilities reduce to the hard 0/1 assignments used by K -means.

(b) (*4 marks*) Describe a short (2–3 step) analysis pipeline you would use to explore structure in a high-dimensional gene-expression dataset (thousands of genes, hundreds of samples) with no known labels, combining a dimension-reduction technique and a clustering technique. State briefly what each step contributes.
